OBJECT-ORIENTED PROGRAMMING / GROUP PROJECT

Hostel Maintenance Request Tracker

PART 1 PROPOSAL • Java + Javalin backend with a simple web front end

THE PROBLEM

When a hostel repair is reported informally (for example, in person or by message), its details can be missed or reported more than once. Residents may also find it hard to know whether a request was received, who is handling it, or whether it is resolved. This tracker gives residents and maintenance staff one shared record for each repair request; these are the problems the project will address, not measured claims about current campus operations.

INTENDED USERS

RESIDENTS

Submit a repair with location and description; check progress by reference number.

MAINTENANCE STAFF

View incoming requests, take responsibility for work, and update progress.

PROPOSED WORKFLOW AND SCOPE

A resident submits a request in a web form. The Java/Javalin backend validates and saves it, then returns a reference number. Maintenance staff view open requests and update status: Received → In Progress → Resolved. Residents check the latest status by reference number. Version one covers request recording, status tracking, and simple persistence (file or SQLite); it excludes emergency dispatch, chat, notifications, and facilities-management integrations.

MAIN CLASSES (6)

CLASS	RESPONSIBILITY
User	Base class for shared identity and contact details; defines the common user role.
Resident	A User who submits repair requests and views their own requests.
MaintenanceStaff	A User who views incoming requests and updates work status.
MaintenanceRequest	Stores reference, location, description, resident, optional assigned staff, status, and timestamps.
RequestService	Validates submissions, coordinates assignment and updates, and enforces valid status changes.
RequestRepository	Saves, finds, and loads requests using one simple persistent store (file or SQLite).

Relationship: Resident and MaintenanceStaff inherit shared identity from User. Both interact with requests through RequestService, which uses RequestRepository for persistence.

TEAM (5 STUDENTS REQUIRED)

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